SERVER:



Reducing Parasitics, Improving Performance : The Case for Low Inductance Interconnects



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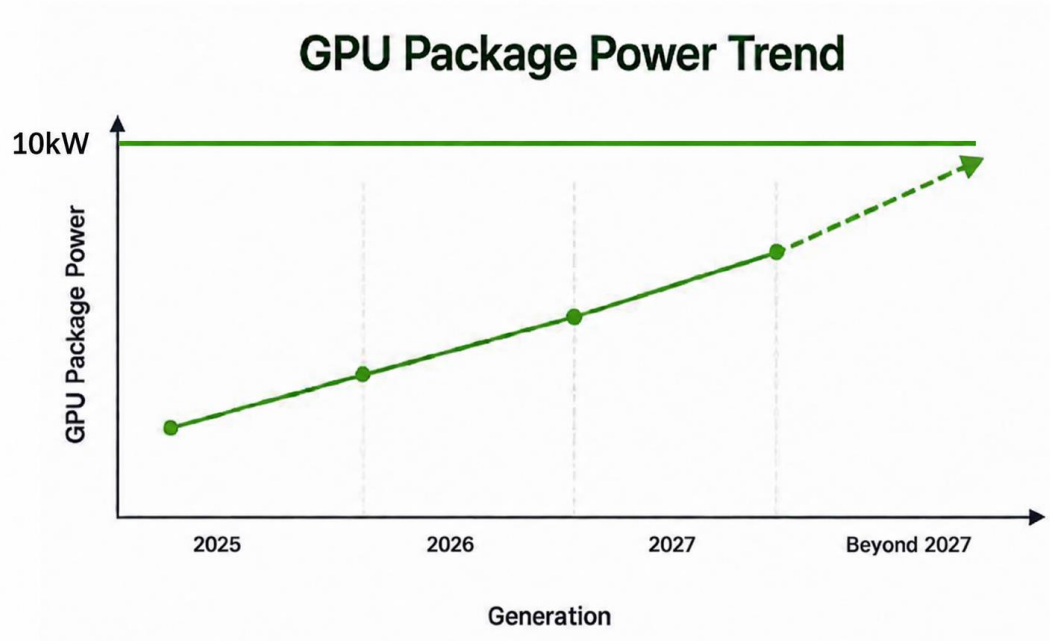
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Co-authors *(TE Connectivity) : Mehdi Mechaik, Ada Zhu, Rupal Pai*

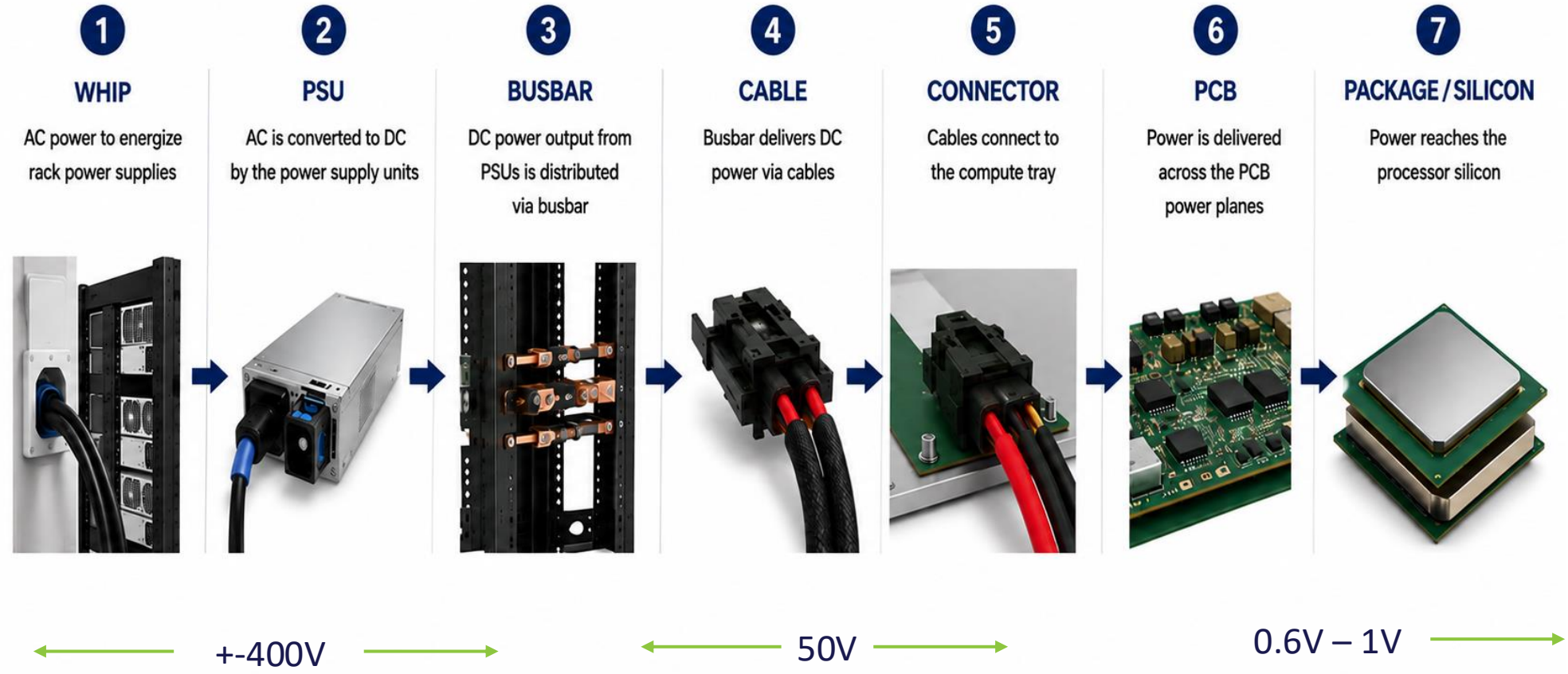
Next Gen AI and HPC Platforms

Transition from Megawatt Data centers to Gigawatt AI Factories

- Adoption of +/-400V or 0-800V for Rack-Level Power
  - Paving the path for 1MW+ Racks
  - Scaling toward Multi-rack POD Architecture
- Growth of Wider, High Power Density AI Trays (e.g. ORW)
- Accelerating GPU Power and Thermal Demands
  - Liquid cooling solutions becoming mainstream
- PDN Becomes a System Level Bottleneck
  - Lower silicon voltages drive exponentially higher current requirements.
  - More power at low voltage means more current.



Power Delivery: FROM WHIP TO CHIP

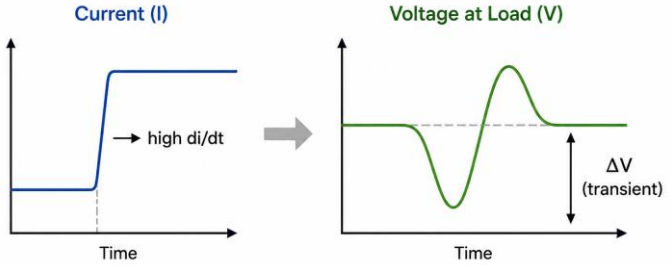


Key Takeaway: Every element in the power path contributes resistance and inductance to the impedance seen by the load.

Clean Power for AI Compute

- Token Economy and Optimizing Power utilization
  - AI racks now consume about 100 kW+.
  - Sensitivity to power events and transients
  - AI workloads create **rapid changes** in processor current

$$\Delta V = \underbrace{I \cdot R}_{\text{DC / Steady-State (ohmic drop)}} + \underbrace{L \frac{di}{dt}}_{\text{Transient (inductive drop)}}$$



- Power Quality no longer just facility requirement
  - Modern GPUs operate at **low core voltages** with tight regulation budgets.
  - Voltage fluctuations reduce power-delivery margins
  - Reduced processing performance and increased timing margin pressure
  - Reduced system stability during **peak workload transitions**

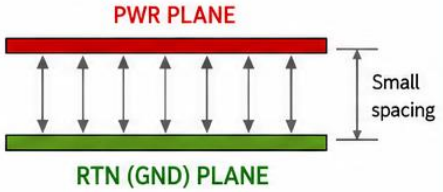
- Resistance and inductance contribute to **voltage disturbance**
  - Delivering **stable voltage**
  - Minimal Voltage droop, overshoot, undershoot (Transient disturbances and noise harmonics)
  - Low core voltages leave little margin for droop and overshoot

Optimizing Inductance

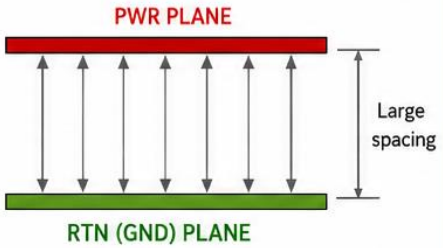
POWER PLANES

Tight coupling minimizes loop area.

GOOD



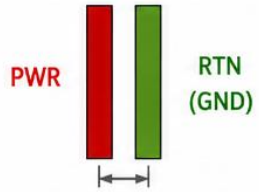
POOR



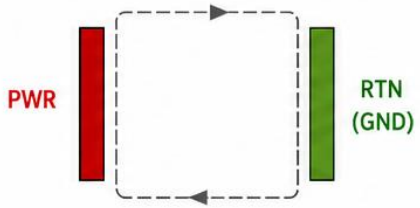
BUSBAR

Closely coupled PWR and RTN conductors minimize loop area.

GOOD



POOR



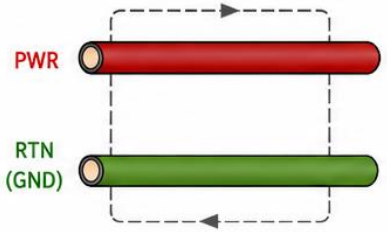
CABLES

Tightly coupled pairs (twisted) reduce loop area.

GOOD



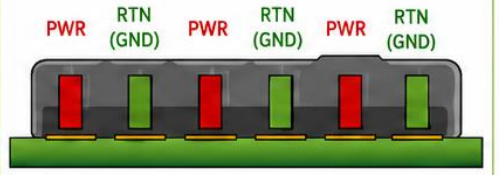
POOR



LEADS / CONNECTORS

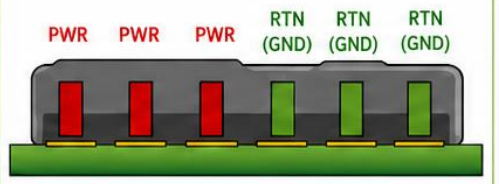
Minimize loop area at the power interface.

GOOD



Closely coupled PWR and RTN paths minimize loop area.

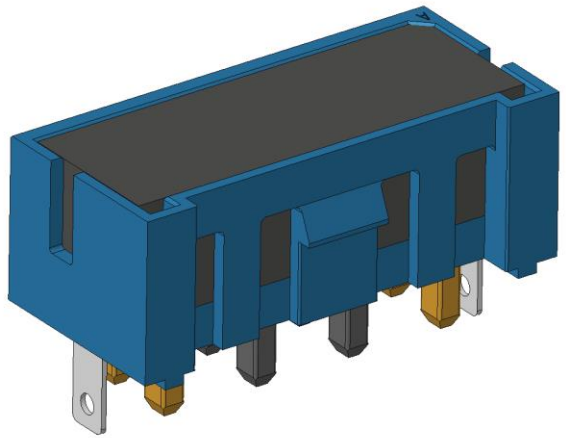
POOR



Separated PWR and RTN paths increase loop area.

Key Takeaway: Minimize loop inductance by keeping power and return paths close together.

Comparable Power Connector



KEY SPECIFICATIONS (Typical)

Current Rating	12.5 A/contact
Contact Pitch	3.0 mm pitch
Contact Resistance	5 mΩ max

BASELINE PERFORMANCE

~8.5 nH

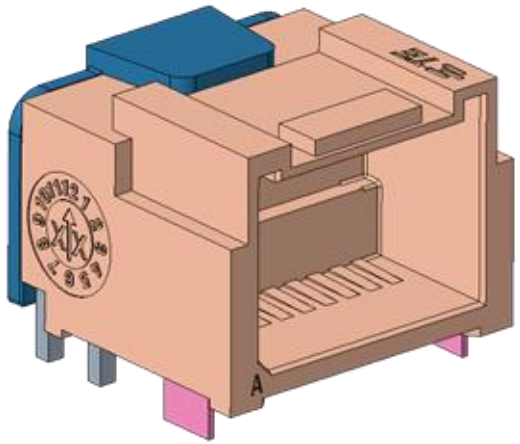
LOOP INDUCTANCE

40 – 100 MHz

Simulation range: 0 – 100 MHz

- A non-optimized connector exhibits ~8.5nH loop inductance.

Low Inductance Connector



DESIGN APPROACH

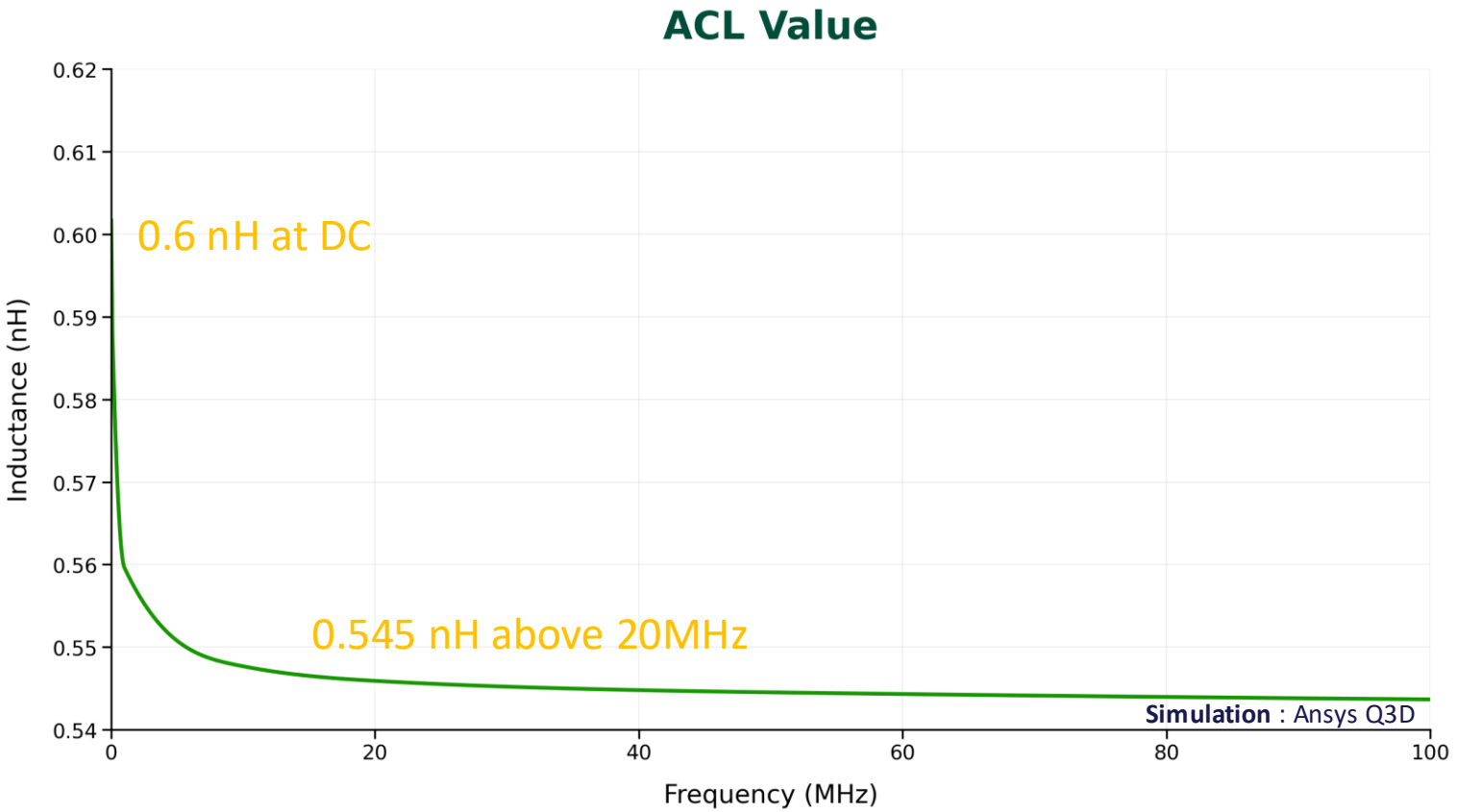
Low-inductance design principles
Keep supply and return paths close to minimize loop area.

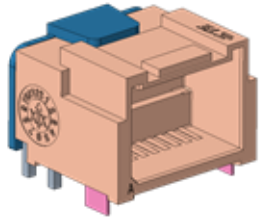
Optimized interconnect geometry
Closely couple PWR and RTN conductors throughout the connector.

Minimized current-loop area
Shorten and constrain the current path through the power interface.

- Connector with optimized parasitic performance.

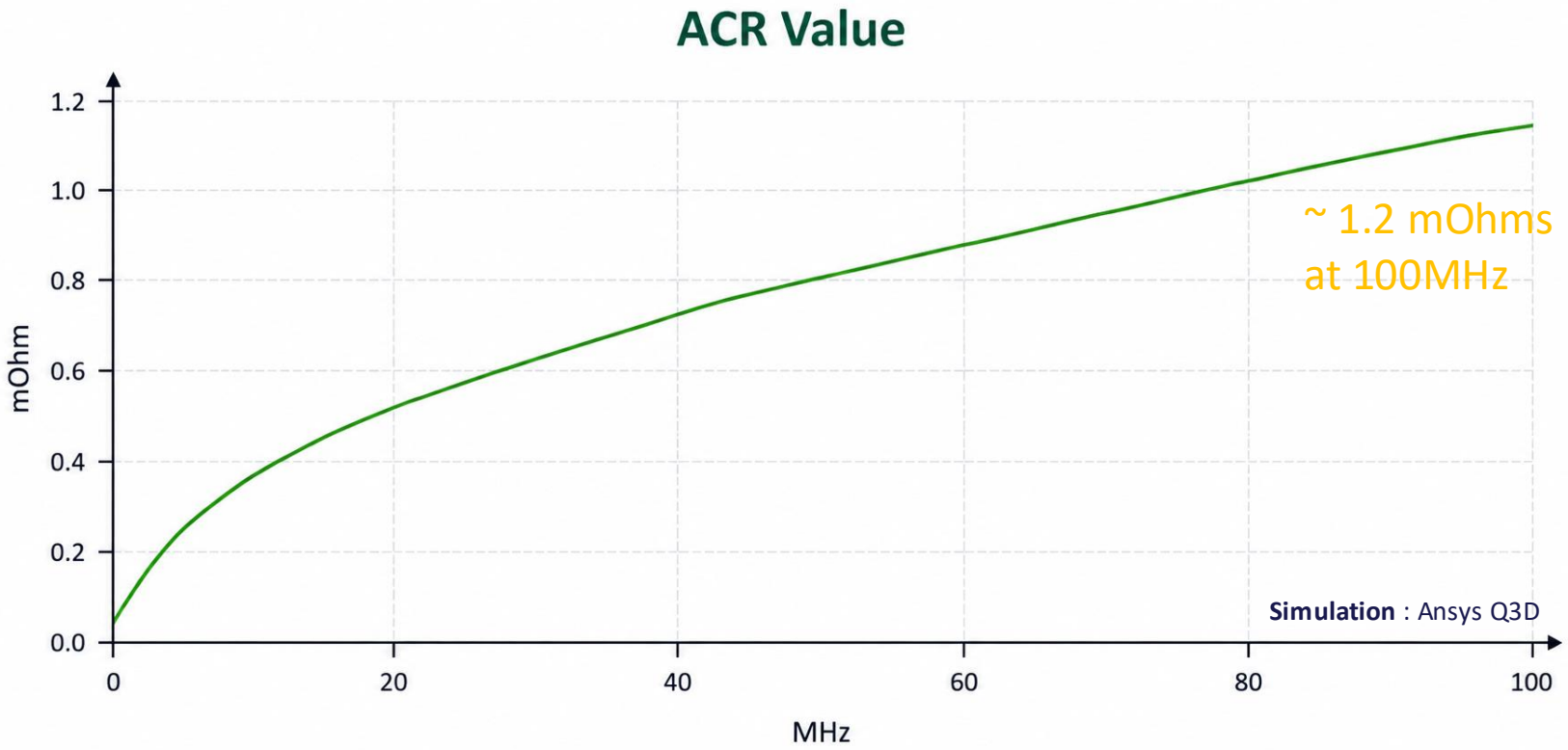
Loop Inductance Plot

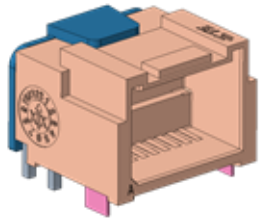




- Inductance stabilizes above 20MHz due to frequency effects.

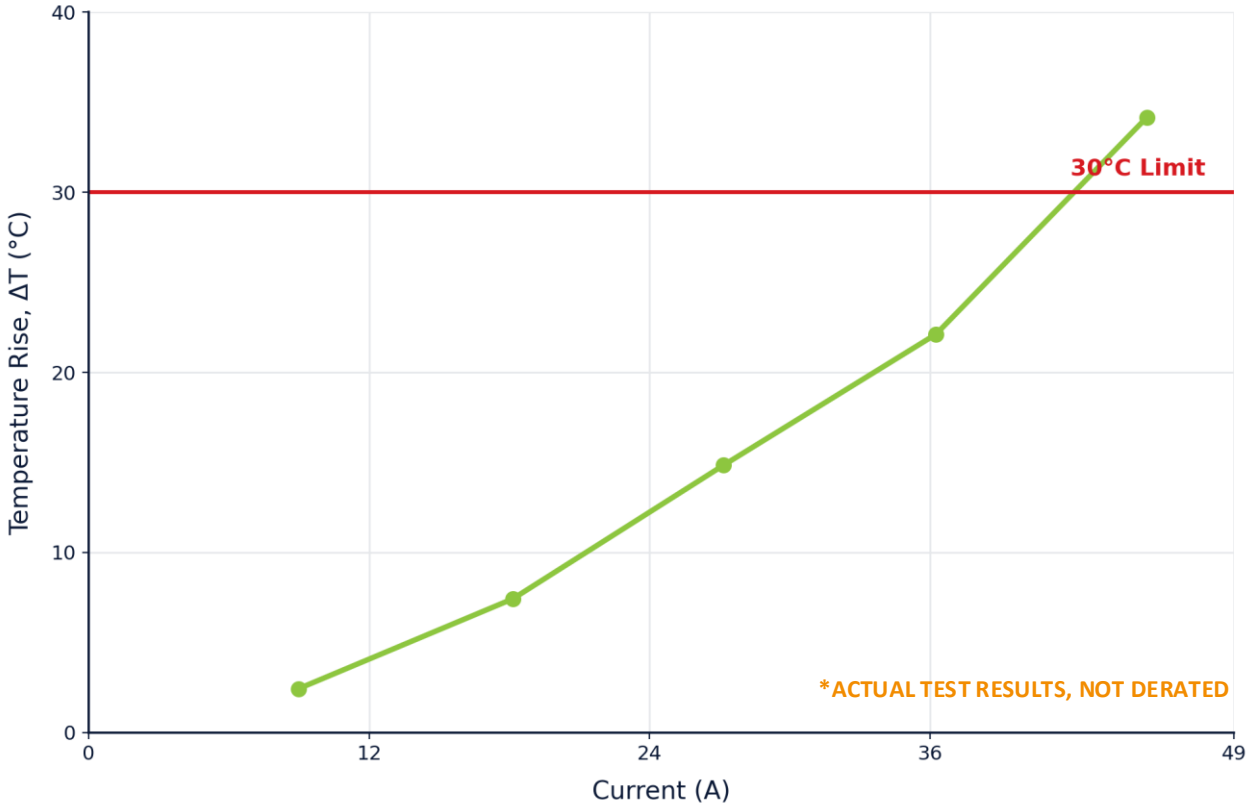
AC Resistance Plot

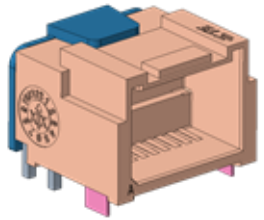




- Resistance keeps increasing due to skin effect.

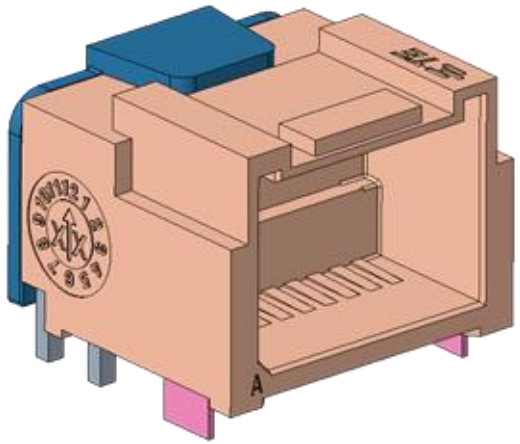
Thermal Performance Test Data





- Supports 34 A/contact within the 30C rise limit.

Optimized Inductance Summary



OPTIMIZED PERFORMANCE

LOOP INDUCTANCE
40–100 MHz

~0.54 nH

~94%
REDUCTION
vs. comparable connector baseline

SUPPORTED CURRENT
34 A/contact

VOLTAGE
12 V

Simulation range: 0–100 MHz

- Geometry optimization reduces loop inductance by ~94%.

Summary

- Rising AI power transients are placing unprecedented demands on PDN performance.
- Rapid current transients generate voltage fluctuations creating power-supply noise.
- Noise increases with faster edge rates and higher switching currents.
- Interconnect inductance is a fundamental design lever governing voltage stability under dynamic loads.
- TE’s optimized interconnect geometry results in lower inductance.

Call to Action

- Characterize and model every power component.
- Validate connector performance through hardware testing.
- Extend low inductance design principles into cable assemblies.
- Contact us at :
  - Tony Howard (tony.howard1@te.com), Quresh Bohra (quresh.bohra@te.com)
  - Product Manager : Kristen Russack (kristen.russack@te.com)
- Where to find additional information:
  - Server Spec
  - Project Wiki

Thank You!



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